

The Physics of Sediment: Fundamental Properties & Statistics



PHYSICAL & MASS PROPERTIES

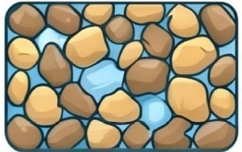
Mineral Composition & Durability



Quartz and feldspar dominate natural sands due to high resistance to mechanical weathering.

Porosity (n) and Consolidation

Poorly Sorted Mixtures (<0.30)



Minimal grain packing

Fresh Clay (>0.8)



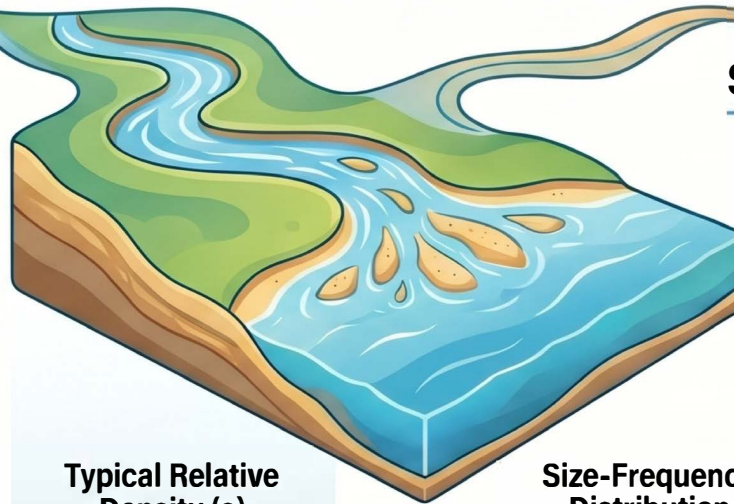
Interstitial voids

Submerged Relative Density (R)



$$R = s - 1$$

Calculated as $R = s - 1$, this value is essential for Shields parameter formulations.



Typical Relative Density (s)



Quartz Sand
~2.65



Limestone
2.6 – 2.8



Magnetite
4.9 – 5.2

GRAIN SIZE & STATISTICAL ANALYSIS

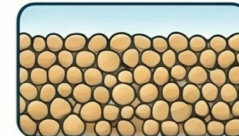


The Phi (φ) Scale



Sorting Parameter (σ_g)

Well-Sorted (<1.3)



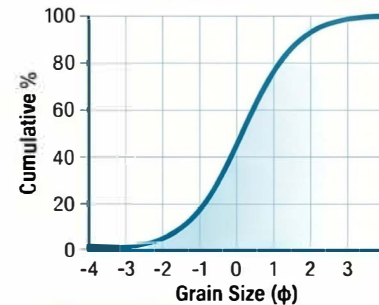
Represents uniformity; values <1.3 are well-sorted.

Poorly Sorted (>1.6)



Size-Frequency Distribution

Natural sediment distributions are often log-normal, best visualized via S-shaped cumulative curves.



Standard Size Classes & Behavior

